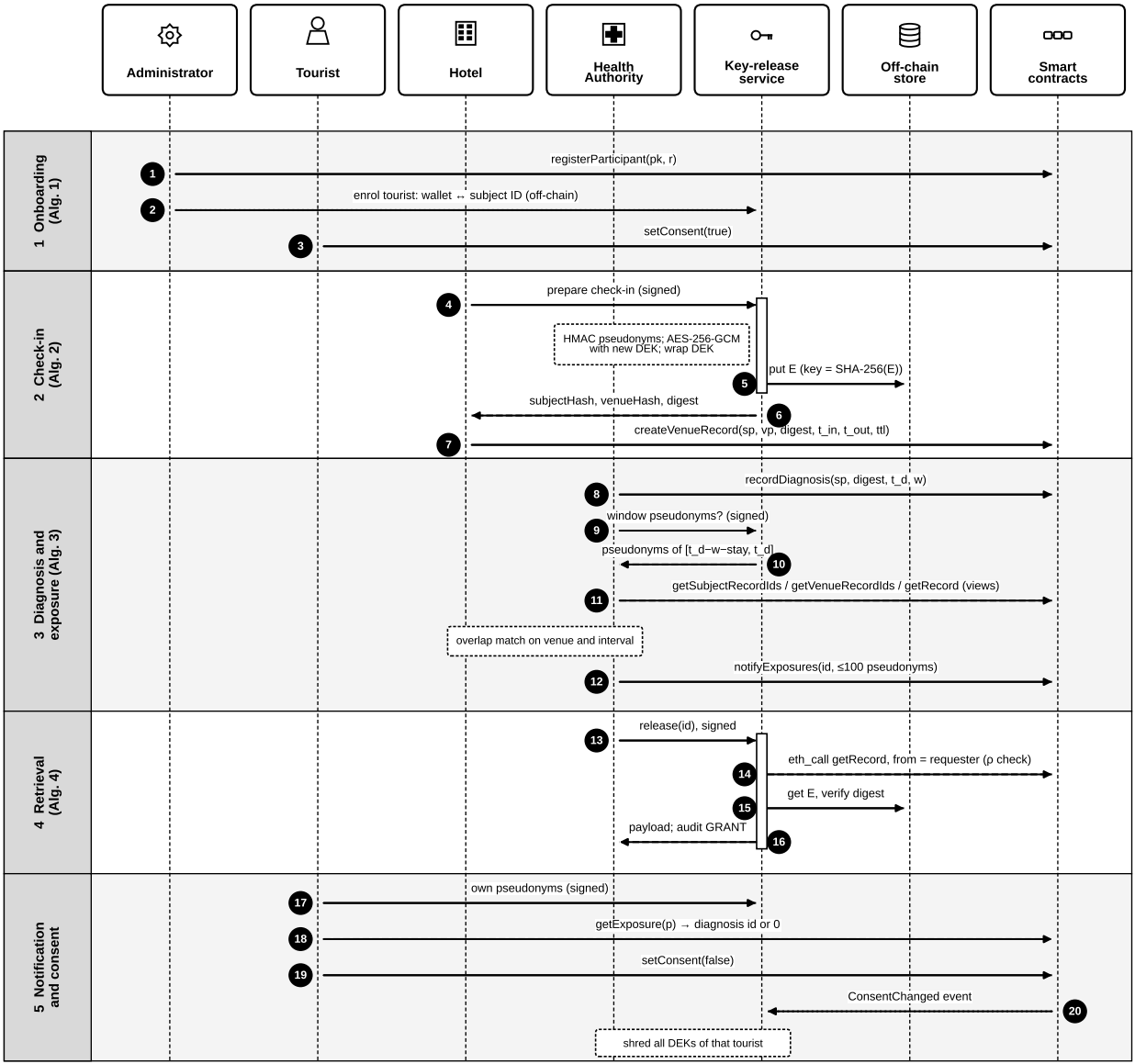
Administrator
Tourist
Hotel
Health Authority
Key-release service
Off-chain store
Smart contracts
1 Onboarding (Alg. 1)
registerParticipant(pk, r)
1
enrol tourist: wallet ↔ subject ID (off-chain)
2
setConsent(true)
3
2 Check-in (Alg. 2)
prepare check-in (signed)
4
HMAC pseudonyms; AES-256-GCM with new DEK; wrap DEK
put E (key = SHA-256(E))
5
subjectHash, venueHash, digest
6
createVenueRecord(sp, vp, digest, t_in, t_out, ttl)
7
3 Diagnosis and exposure (Alg. 3)
recordDiagnosis(sp, digest, t_d, w)
8
window pseudonyms? (signed)
9
pseudonyms of [t_d−w−stay, t_d]
10
getSubjectRecordIds / getVenueRecordIds / getRecord (views)
11
overlap match on venue and interval
notifyExposures(id, ≤100 pseudonyms)
12
4 Retrieval (Alg. 4)
release(id), signed
13
eth_call getRecord, from = requester (p check)
14
get E, verify digest
15
payload; audit GRANT
16
5 Notification and consent
own pseudonyms (signed)
17
getExposure(p) → diagnosis id or 0
18
setConsent(false)
19
ConsentChanged event
20
shred all DEKs of that tourist